Environmental Engineering

Unit III: Water Storage and Distribution

Syllabus

Storage and balancing reservoirs - Types, location and capacity. Distribution system: layout, hydraulics of pipe lines, pipe fittings, valves including check and pressure reducing valves, meters, analysis of distribution systems, leak detection, maintenance of distribution systems, pumping stations and their operations - House service connections.

1. Storage and Balancing Reservoirs

Treated water is stored in **service (distribution) reservoirs** so that a uniform rate of treatment and pumping can meet the **fluctuating demand** of the town. They are placed between the treatment plant and the distribution system.

Functions of service reservoirs

- **Balance** the variation between supply (constant) and demand (variable)
- Provide a **reserve** for fire fighting and emergencies (breakdown of pumps or mains)
- Maintain **pressure** in the distribution system
- Allow pumps to work at a **uniform and efficient rate**, which cuts the cost of pumping
- Reduce the size of the **trunk main** (designed for average, not peak demand)
- Give time for **chlorine contact**

Types of reservoirs

Type	Description
Underground (ground level) reservoir	RCC / masonry tank at ground or below it; for flat towns, when natural elevation is not available; needs pumping to the system
Elevated reservoir (overhead tank)	Tank on a tower, for flat terrain; water flows by gravity; supports pressure; capacity small (up to a few lakh litres)
Surface (ground) reservoir on a hill	Natural high ground used, gravity supply
Standpipe	Tall cylindrical tank, upper part gives effective storage (balancing)
Balancing reservoir	Stores surplus during low demand and gives out during peaks
Storage reservoir	Large capacity, stores monsoon flow

Shapes and materials

- **Rectangular** - small capacity; **circular** - more economical in walls; **Intze type** - elevated tanks with conical dome bottom
- Materials: RCC (most common), steel (elevated), masonry, prestressed concrete, **GRP / HDPE** for small tanks
- Needs **roof cover**, ventilation, manhole, overflow, scour and outlet pipe; depth 3-6 m; **free board** 0.5 m, and floor slopes to a sump

Location and height of reservoirs

- Should be **centrally located** near the demand centre, at a high place to give gravity pressure
- Height decided to give the residual pressure of 7-15 m at the farthest point, with friction loss; **minimum 12-15 m above the ground** for a 3-storey building
- Elevated reservoir connected to the distribution through **inlet and outlet** (separate or same pipe)
- **Dead storage** (minimum level) never used

Capacity of a service reservoir

- Capacity = **balancing storage + breakdown storage + fire storage**
- **Balancing storage** is about 15-20% of the daily demand (or found from mass curve method)
- **Breakdown storage** - 1 / 4 of the total daily demand (for emergencies like pump failure)
- **Fire storage** - based on the fire demand for 4 hours (e.g., 1 lakh litres per hour for large towns)
- Total capacity is commonly **one third to one-half of the daily demand** (8 - 12 h)

Mass curve (analytical method) of balancing storage

Plot cumulative demand vs time (hours) → Draw supply line: cumulative supply (uniform rate) → Draw parallel lines tangent at highest and lowest points of the demand curve → Vertical distance between the tangents = **balancing capacity**

Example

Daily demand 4 MLD. Pump 24 h; max surplus = 8% and max deficit = 10% of daily demand. Balancing storage = (8 + 10)% x 4 = 0.72 ML. Add 25% breakdown (1 ML) + fire (say 0.2 ML) => total = **1.92 ML**.

2. Distribution System

The **distribution system** carries treated water from service reservoirs to the consumers in sufficient quantity, at **adequate pressure**, without deterioration of quality. It accounts for **40–70% of the cost** of the water supply scheme.

Requirements of a good distribution system

- Water to be supplied at all points with **required pressure** (min. 7–10 m, 12 m for 3-storey)
- Quantity adequate for domestic and **fire-fighting** demands
- **Water quality** should not deteriorate in pipes
- Easy **maintenance**, repair and extension; **water-tight** against leaks and entry of contaminants
- Layout with **minimum dead ends** and possible isolation of sections with valves

Methods of distribution

Method	Detail	Merits / Demerits
Gravity system	Source or reservoir above the town; no pumps	Cheapest, reliable. Suitable for hilly places
Pumping system	Direct pumping to mains	Pressure fluctuates, power failure stops supply
Combined (dual) system	Pumping to elevated tank, and gravity to the town; excess stored	**Most economical and reliable**

Systems of supply

- **Continuous system** – supply for 24 h, fresh water, good pressure for fire; but more wastage
- **Intermittent system** – supply for some hours (common in India), smaller pipes; but **risk of contamination** (negative pressure backflow), water stored in houses, pipe corrosion

3. Layout of Distribution Systems

Layout	Description	Advantages / Disadvantages
Dead end (tree) system	One main with sub-mains and branches; ends are dead	Simple, cheap, easy design; **stagnation at dead ends**, pressure drops if a part is repaired, poor fire flow
Grid iron system	Interconnected mains and branches in parallel (rectangular)	Free circulation, no dead ends, water available from many directions during repairs; **costly**, difficult analysis, many valves
Ring (circular) system	Main pipe round the town / area with branches	Good for fire; flow from both sides; needs more pipe length
Radial system	Zones from a central reservoir, flow radially outward	Quick, high pressure, simple design; for zoned towns

Choice of layout

* Depends on **town plan**, topography, street pattern and cost
* Grid iron or combined system for planned cities; dead-end for old and irregular towns
* Large cities use **zoning** (several pressure zones) from separate reservoirs

4. Hydraulics of Pipe Lines

Flow in pipes is turbulent. Head loss due to friction is found by **empirical formulae**; the pipe is designed for the required discharge and pressure.

Head loss formulae

Darcy-Weisbach: $h_f = f\,L\,v^2 / (2\,g\,D)$ (f = friction factor = 64/Re for laminar flow)

Hazen-Williams: $v = 0.849\,C\,R^{0.63}\,S^{0.54}$; $h_f = 10.67\,L\,Q^{1.852} / (C^{1.852}\,D^{4.87})$ [C = 100-140]

Manning's: $v = (1/n)\,R^{2/3}\,S^{1/2}$ (for large pipes flowing full: $R = D/4$)

- Common C values: new cast iron 130, old CI 100, PVC / plastic 150, concrete 120, steel 120
- **Minor losses** - entry, exit, bends, valves = $K\,v^2/2g$, small in long pipes
- **Velocity in mains** 0.6 - 2.0 m/s (minimum to prevent silting, maximum to limit loss and water hammer)
- **Pipes in series** - same Q, losses add; **pipes in parallel** - same head loss, flow splits
- **Equivalent pipe** - Dupuit formula: $L/D^5 = L_1/D_1^5 + L_2/D_2^5 + \ldots$

- **Water hammer** - sudden pressure rise on rapid closure of valve; Joukowsky: delta p = rho c v; prevent by slow valves, surge tanks, air vessels, relief valves

Pipe materials for distribution

Material	Merits	Demerits
Cast iron / ductile iron	Strong, long life (100 yrs), easy to join and tap	Heavy, tuberculation, corrodes
Mild steel / GI	Light, high pressure, easy to weld	Rusting, needs coating; GI small diameter (service)
PVC / uPVC / HDPE	Light, smooth, corrosion free, cheap	Low strength, affected by heat, UV
RCC / PSC	Large diameter, durable, low friction	Heavy, difficult to repair
Asbestos cement	Light, cheap	Brittle, health hazard (not used now)
Copper / lead	House service	Costly (lead is toxic)

5. Pipe Fittings

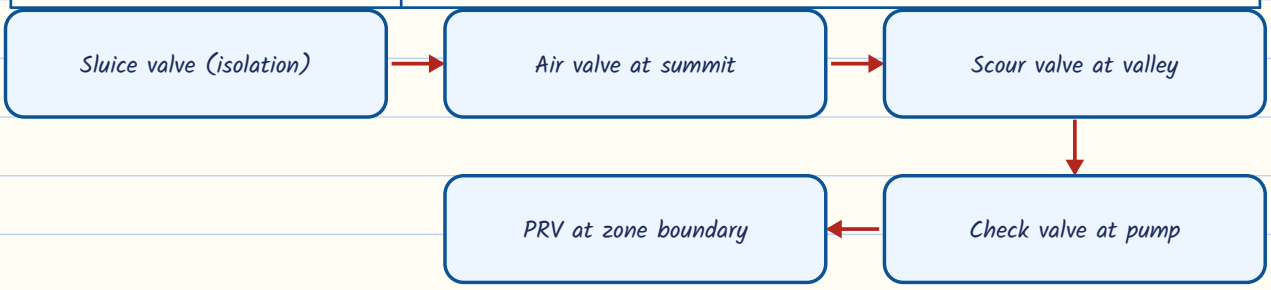
- **Sockets / couplings** - join two straight pipes
- **Bends, elbows (90, 45 degrees), tees (T), crosses, reducers / increasers** - change direction or diameter
- **Unions, flanges, nipples** - for easy removal
- **Plugs and caps** - closing ends; **bell-mouth**, **bib cocks**, **stop cocks**
- **Thrust blocks** (concrete anchor) at bends and tees to take hydraulic thrust
- **Expansion joints and flexible joints** (to take movement) and **specials**

Types of pipe joints

Joint	Used for
Spigot and socket (bell and spigot) - lead / rubber ring	CI pipes, most common
Flanged joint - bolts and gasket	Pumping stations, valve connections
Flexible (tyton) joint with rubber gasket	DI pipes, takes ground movement
Expansion joint	Long pipes in temperature change
Solvent cement joint	PVC pipes
Welded joint	Steel pipes

6. Valves

Valve	Function
Sluice (gate) valve	Isolate sections, open or close fully; installed at every 150-300 m and each branch
Check (reflux / non-return) valve	Allows flow in **one direction only**, prevents backflow; fixed on pump delivery, to protect the pump from reverse flow and water hammer
Pressure reducing valve (PRV)	Reduces and keeps a **constant downstream pressure** irrespective of upstream; for pressure zones; avoids bursts and leakage
Air relief (air) valve	Releases air accumulated at **summits** (high points) and admits air during emptying
Scour (blow-off) valve	At low points; drains the main and removes silt
Fire hydrant	Outlet for firefighting, at 100-150 m spacing; pressure 1-1.5 kg/cm^2
Altitude valve	Closes inlet when the elevated tank is full
Butterfly valve	Large mains, quick control, compact
Float valve, pressure relief valve	Controls water level / relieves excess pressure



7. Water Meters

Water meters measure the quantity of water consumed. They help in revenue collection, **reduce wastage** (up to 20-30%) and are used for finding leaks and for analysing the demand.

Type	Working
Positive displacement (volumetric) meter	Measures volume in each cycle (piston, oscillating disc); accurate at low flow; domestic
Velocity (inferential) meter	Turbine / propeller rotates with speed of flow, for large flows; (Woltmann type)
Electromagnetic / ultrasonic	No moving parts, accurate; used in large mains
Venturi meter / orifice meter	Pressure difference gives discharge; used in treatment plants and mains

Type	Working
Domestic meter / bulk meter	Small (15 mm) vs. large mains meter

- Meters need **strainer, regular testing and calibration**; installed in a meter chamber at a convenient location
- Advantages: equitable tariff, detects leakage, **discourages waste**; disadvantages: cost, maintenance, errors at low flow

8. Analysis of Distribution Systems

Analysis finds the **flows in the pipes and pressures at the nodes** of a network for a given demand. It is done to design pipe sizes and check the residual pressures.

Methods

Method	Use
Equivalent pipe method	Replace a part of network by single equivalent pipe, simple
Circle method	For a circular (ring) layout
Electrical analogy	Analog computer (Mc Ilroy analyser)
Hardy Cross method	Most common, iterative, for loops
Newton-Raphson, linear theory, computer packages	EPANET, WaterGEMS, LOOP, for large networks

Hardy Cross method (method of balancing heads)

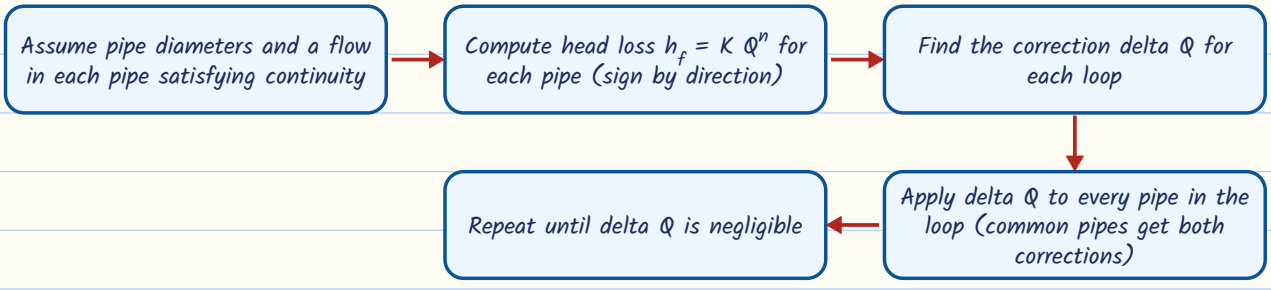
Principles

1. At every node, **inflow = outflow** (continuity): sum of $Q = 0$

2. In every closed loop, **algebraic sum of head losses = 0**: sum of $h_f = 0$ (clockwise +ve, anticlockwise -ve)

$h_f = K Q^n$; $n = 1.85$ (Hazen-Williams) or 2 (Darcy-Weisbach)

Correction: $\Delta Q = -\text{sum}(K Q^n) / \text{sum}(n K Q^{n-1}) = -(\text{sum } h_f) / (n \text{ sum}(h_f/Q))$



- Calculated pressure at nodes = elevation of reservoir − head loss − ground level; **residual pressure** at critical node must be at least 7 – 15 m
- **Design for peak demand** (maximum hourly) and check for maximum daily demand + fire flow
- Maximum **fire demand condition**: critical point gets the fire flow plus normal demand

9. Leak Detection and Water Losses

Unaccounted-for-water (UFW) in Indian cities is 30–50%. Losses occur from leaking pipes, joints, valves, house connections, illegal taps, and meter errors.

Methods of leak detection

- **Water audit** and **district metering areas (DMA)** – measure inflow in a district at night (minimum night flow) to detect leaks
- **Sounding (listening) rod / geophone** – hear the hissing sound of leak on valves and hydrants
- **Leak noise correlators** – two sensors on pipe, find the leak position from time delay
- **Ground microphones, acoustic loggers, tracer gas (helium, hydrogen), infrared thermography, ground-penetrating radar**
- **Pressure testing** of pipelines, **visual observation** of wet patches or fall in pressure
- **Control of leakage:** pressure management using PRVs, repair of leaks at once, replacement of old pipes, good quality fittings, regular meter calibration and quick service-connection repair
- **Active leakage control** programme with trained teams, **GIS and SCADA** for network monitoring

10. Maintenance of Distribution Systems

- **Periodic flushing** of mains by opening scour valves and hydrants, to remove silt and sediment
- **Swabbing and cleaning** of pipes (pigging) to remove tuberculation, followed by cement-mortar lining
- **Disinfection of new and repaired mains** with chlorine (50 mg/L for 24 h, or 20 mg/L for 24 h) before commissioning
- **Valve exercising and inspection** every year; hydrants maintained

- Maintain **residual chlorine** (0.2 mg/L at the end); sample water at various points to check bacteriology
- Prevention of **cross connection** (sewer and water pipes: water must be above sewer with 3 m clear distance)
- **Corrosion control:** lining, pH adjustment, cathodic protection
- **Records, maps** and GIS of the pipe network, repair crews, and public complaint system

11. Pumping Stations and Their Operations

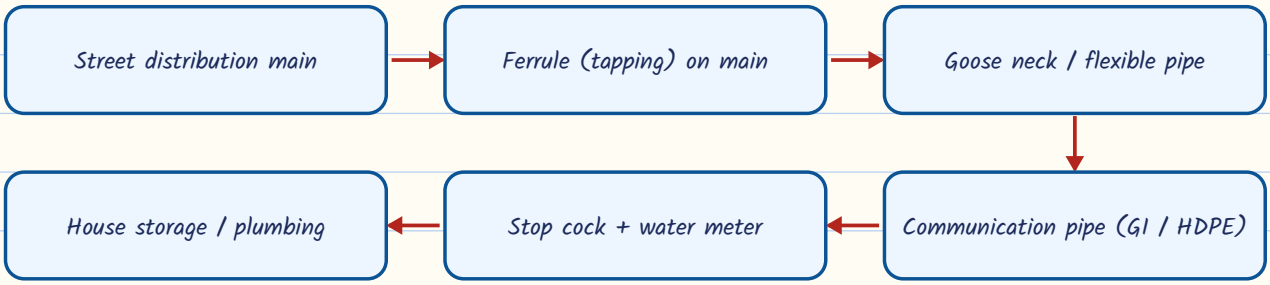
Pumping stations lift water from the source (raw water), through the treatment plant (intermediate), and to the distribution system (**booster** and **high-lift pumping**).

Pump type	Notes
Centrifugal pump	Most common; impeller, volute casing; high discharge, smooth flow; needs priming (except self-priming and submersible)
Reciprocating pump	Piston; high head, low discharge; pulsating flow
Submersible pump	Motor and pump together under water, for deep tubewells
Axial and mixed flow pump	Low head and high discharge
Jet pumps, air lift pumps, hydraulic ram	Special

- **Components:** suction well (sump), pumps and motors, delivery pipes, **foot valve and strainer**, check valve, sluice valve, priming device, control panel, **surge protection**, standby units (one standby per 3-4 working pumps)
- **Operation:** priming, start with delivery valve closed (for centrifugal), then open slowly; record discharge, head, current, voltage, bearing temperature and vibration; **log book**
- **Maintenance:** lubrication, replacement of seals and bearings, alignment, cleaning of strainer, painting, testing of motors, **preventive maintenance schedule**
- **Pump efficiency** = water power / input power; Water power (kW) = gamma Q H / 1000, 70-85% overall efficiency
- **Selection** by head-discharge curve, NPSH (available > required, to avoid **cavitation**), speed and cost

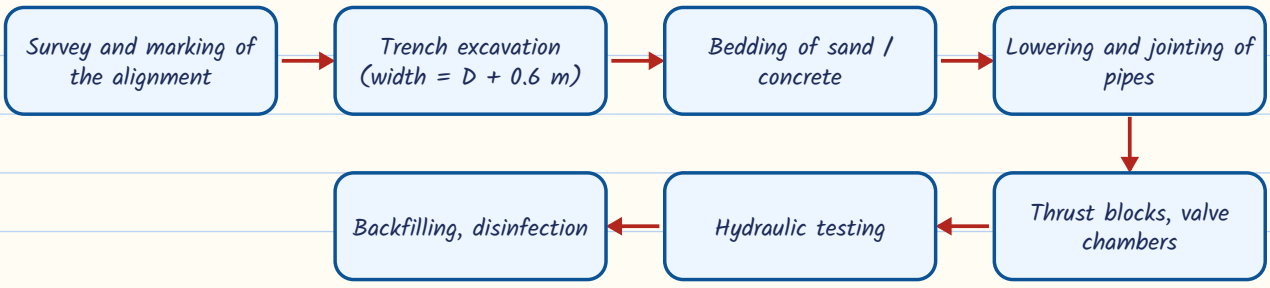
12. House Service Connections

A **service connection** is the pipe arrangement that connects the **street main** with the house plumbing.



- **Ferrule** is screwed into the main by a tapping machine under pressure
- **Goose neck** (lead, copper or PE) gives flexibility against settlement
- **Communication pipe** - from main to the property boundary, up to the stop cock; **service pipe** - inside the premises
- Diameter 12 - 25 mm; **stop cock** in a pit near the boundary; **meter** installed after the stop cock
- Connection pipe should not be near drains; avoid **back siphonage** (use non-return valves, break tank) to prevent contamination of the main
- Material: GI, MDPE/HDPE, copper (earlier lead is banned)

13. Laying and Testing of Pipes



- **Depth of cover** - minimum 1 m (0.9-1.2 m) below the road to avoid damage from traffic loads and frost
- **Water mains are laid above sewers**, with minimum 3 m horizontal separation, and on the **opposite side** of the street from drains where possible

- Pipes laid in straight lines and uniform gradients, with **air valves at summits and scour valves at valleys**
- **Hydrostatic test** - fill the section with water, pressurise to **1.5 times the working pressure** (or as specified, usually 2-3 kg/cm^2 above), hold for 1-2 h; leakage should not exceed permissible limits
- **Bacteriological test** after disinfection; flush before commissioning

Corrosion of pipes and its control

- **Causes:** acidic or soft water, dissolved oxygen, differential aeration, galvanic action between dissimilar metals, stray currents, bacteria (sulphate reducing, iron)
- **Tuberculation** - build-up of rust nodules, reduces the carrying capacity (C value falls from 130 to 90)
- **Control:** cement-mortar or epoxy lining, bitumen coating, **pH and alkalinity control** (lime, soda ash), **corrosion inhibitors** (polyphosphates, silicates), **cathodic protection** (sacrificial anodes), avoid dissimilar metals

14. Design Problems (solved examples)

Example 1: Capacity of an elevated tank

Population 30,000 at 135 lpcd => demand = 4.05 ML/day. If pumped only 8 h/day, the pumping rate = 4.05/8 = 0.506 ML/h (continuous 24 h pumping gives 0.169 ML/h).

Take balancing storage = 33% of daily demand = 1.34 ML.

Add fire storage 0.2 ML => total capacity = **1.54 ML** (about 1.6 ML).

Diameter of circular tank (depth 4 m): A = 1540/4 = 385 m^2, so D = 22.1 m.

Example 2: Head loss in a pipe (Hazen-Williams)

L = 1000 m, D = 300 mm, Q = 0.1 m^3/s, C = 100.

$h_f = 10.67 \times 1000 \times (0.1)^{1.852} / (100^{1.852} \times 0.3^{4.87})$

$= 10.67 \times 1000 \times 0.01406 / (5058 \times 0.00285) = 150 / 14.4 =$ **10.4 m**

Velocity v = Q/A = 0.1/0.0707 = 1.41 m/s (within 0.6-2 m/s, OK).

Example 3: Hardy Cross, first iteration

Loop ABCD: Q_{AB} = 40 L/s (K = 2.0), Q_{BC} = 15 (K = 3.0), Q_{AD} = 25 (K = 3.5), Q_{DC} = 10 (K = 2.5) in the clockwise sense (AB, BC clockwise; AD, DC anticlockwise).

$h_f = K Q^2$ (in suitable units): AB = +3.2, BC = +0.68, AD = -2.19, DC = -0.25 => sum h_f = 1.44

sum(2 K Q) = 0.475; delta Q = -1.44/0.475 = **-3.0 L/s** (reduce clockwise flows by 3.0, increase anticlockwise by 3.0) and repeat.

Pressure zones and distribution in hilly towns

- Pressure in a pipe should not exceed **60–70 m** (6–7 kg/cm^2) in town; high pressure leads to bursts, high leakage
- For tall differences in elevation, the town is divided into **zones**, with a separate reservoir or PRV for each; a **break pressure tank** is used in long gravity mains in hills
- **Dual supply / interconnection** between zones is provided for emergencies

Remember for exam

* Capacity of service reservoir = balancing + breakdown + fire storage
* Dual system (pumping + gravity) is the most economical; grid iron layout has no dead ends
* Hardy Cross: sum of Q at node = 0, sum of head losses in a loop = 0
* Hazen-Williams C: new CI = 130; PVC = 150; velocity 0.6 – 2 m/s
* Check valve = no backflow; PRV = constant downstream pressure; air valve at summits; scour valve at low points
* Hydrants at 100–150 m; minimum residual pressure 7–15 m

Important questions

1. Explain the types, location and capacity of service reservoirs. How is the balancing storage determined?
2. Describe the different layouts of distribution systems with merits and demerits.
3. Explain the methods of distribution: gravity, pumping and dual systems.
4. Explain the various valves used in distribution systems with their functions.
5. Describe the Hardy Cross method for analysing a pipe network.
6. Write notes on water meters and leak detection methods.
7. Discuss the maintenance of distribution systems.
8. Explain the working and operation of a pumping station.
9. Describe the house service connection with a neat sketch.